Pranav Achar

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Robot learning, computer vision and AI agents. Seeking Summer 2027 software and machine learning internships.

Education

Los Angeles Mission College **Expected transfer: Fall 2027**
A.S. Computer Science & Mathematics (transfer track) 2026 to present
4.0 GPA at LA Mission College. Also coursework at Diablo Valley College and Coastline College.
Coursework: Calculus I and II, Programming in C++; *in progress:* Calculus III, Advanced C/C++; *Spring 2027:* Linear Algebra & Differential Equations, Discrete Mathematics, Data Structures.

Experience

AI Research Intern *UC Santa Cruz AIEA Lab* multi-step logical reasoning in LLMs May 2026 to present
- Built a harness that runs three 4-bit LLMs (Qwen2.5 1.5B and 3B, Llama 3.2 3B) locally on Proof Writer, GSM8K and FOLIO and checks every stated claim and calculation against ground truth (Python, MLX).
- Found that 63 to 72% of wrong GSM8K answers had correct arithmetic on the wrong plan, and that a verifier which removed every checkable step error still did not beat plain chain-of-thought for either Qwen model (Python, MLX).

Engineering Intern *Opterra Energy Solutions* Jun 2026 to Aug 2026
- Took on an open engineering proposal project: designed a Heat-Recovery-as-a-Service system that sells AI data center waste heat as district heat, and built its financial model (about $3.5M/yr per 10 MW site, 2 to 4 year payback).
- Pitched it in a full deck with engineering schematics (Blender, Three.js).

Independent Researcher *Vision-language models (VLMs) for camera-based event monitoring* Jul 2026 to present
- Researching VLMs that run on the camera device, notice and flag observable events in video (a fall, a person down) and save reviewable clips; VLM work in progress (PyTorch, RT-DETR, ONNX).
- Trained a pose-based prototype fall model over 100+ GPU hours; on a public held-out benchmark it never trained on (UR Fall), it flagged all 14 scored falls a median 2.6 s after onset (PyTorch, RTMPose, ONNX).

Freelance Web Developer *Clients: Peninsula Tamil School, Anushma LLC, CPFA* 2024 to 2026
- Rebuilt the Peninsula Tamil School site as a bilingual English and Tamil app with WCAG 2.1 AA fixes, cutting load time from 4.2 s to 1.8 s and raising Lighthouse from 68 to 98 (Next.js, React, Vercel).
- Built a 150+ product catalog site with quote requests for lab-materials distributor Anushma LLC (React, Vanta.js) and a site for the cybersecurity nonprofit CPFA (JavaScript, Netlify).

Open Source Developer *statsmodels* Jul 2026 to present
- Contributed to statsmodels, the standard Python statistics library (11.7k GitHub stars, used by 181k repositories): 11 bug fixes, two of them for issues open since 2018, plus GLM documentation (Python, NumPy, pytest).

Projects and Awards

Tangible | **2nd Place, Apify Track**, Cal Hacks 13.0 [code] [live] Oct 2026
- Built agents that turn a one-line device description into a 3D build from real open-source parts, with a case, a KiCad board that passes DRC and firmware; an Apify Actor prices the parts (Python, Claude, KiCad, Apify).

Etch | **1st Place**, Recursive Agents Hackathon ($14,000 prize) [code] [live] Oct 2026
- Built Nemotron agents that turn a spoken request into a routed KiCad board of real parts, checked by ERC, DRC and an ngspice simulation, with up to three repair tries per board (Python, KiCad, ngspice).

Understudy | **1st Place**, Runway API Hackathon ($10,000 prize) [code] [live] Sep 2026
- Turns one sentence into Runway-generated, physics-checked robot demos and trains a SmolVLA policy on a rented GPU, scoring 43/50 vs 44/50 for perfect scripted demos in simulation (Python, Runway API, LeRobot, MuJoCo).

Player Two | **1st Place**, Executable World AI Hackathon; **2nd Place**, Battle of the Personal Brains [code] [live] Sep 2026
- Built an agent that turns a plain-language task into robot training data: it finds Creative Commons video, tracks pose and retargets it onto Unitree G1, SO-101 and Franka Panda robots (Python, MediaPipe, MuJoCo).
- One unattended run screened 68 videos in under 17 minutes and exported the 22 that passed deterministic checks as a LeRobot v3.0 dataset (LeRobot, AWS Strands, Cognee, Docker).

More wins
- **SixthSense** [code], 2nd Place, ExecuTorch Hackathon (Qualcomm x Meta): offline navigation copilot for blind users, featured on the PyTorch blog. **Foundry** [code], 1st Place, Superserve Track, terac One-Person-Company Hackathon: agent that builds a storefront per person (Next.js, Claude). **HealthFlow** [code], 1st Place, Scalekit x Apify: 9-agent voice-to-EHR pipeline (TypeScript, LangChain). **1st Place**, Apify Track, Build Club.

Skills

Languages: Python, TypeScript, JavaScript, C++, Kotlin, Java, SQL, Rust
ML and robotics: PyTorch, ExecuTorch, MLX, MediaPipe, RTMPose, YOLO, RT-DETR, ONNX, MuJoCo, LeRobot
Agents and systems: Anthropic SDK (Claude), LangGraph, LangChain, AWS Strands, Next.js, React, Postgres, Docker